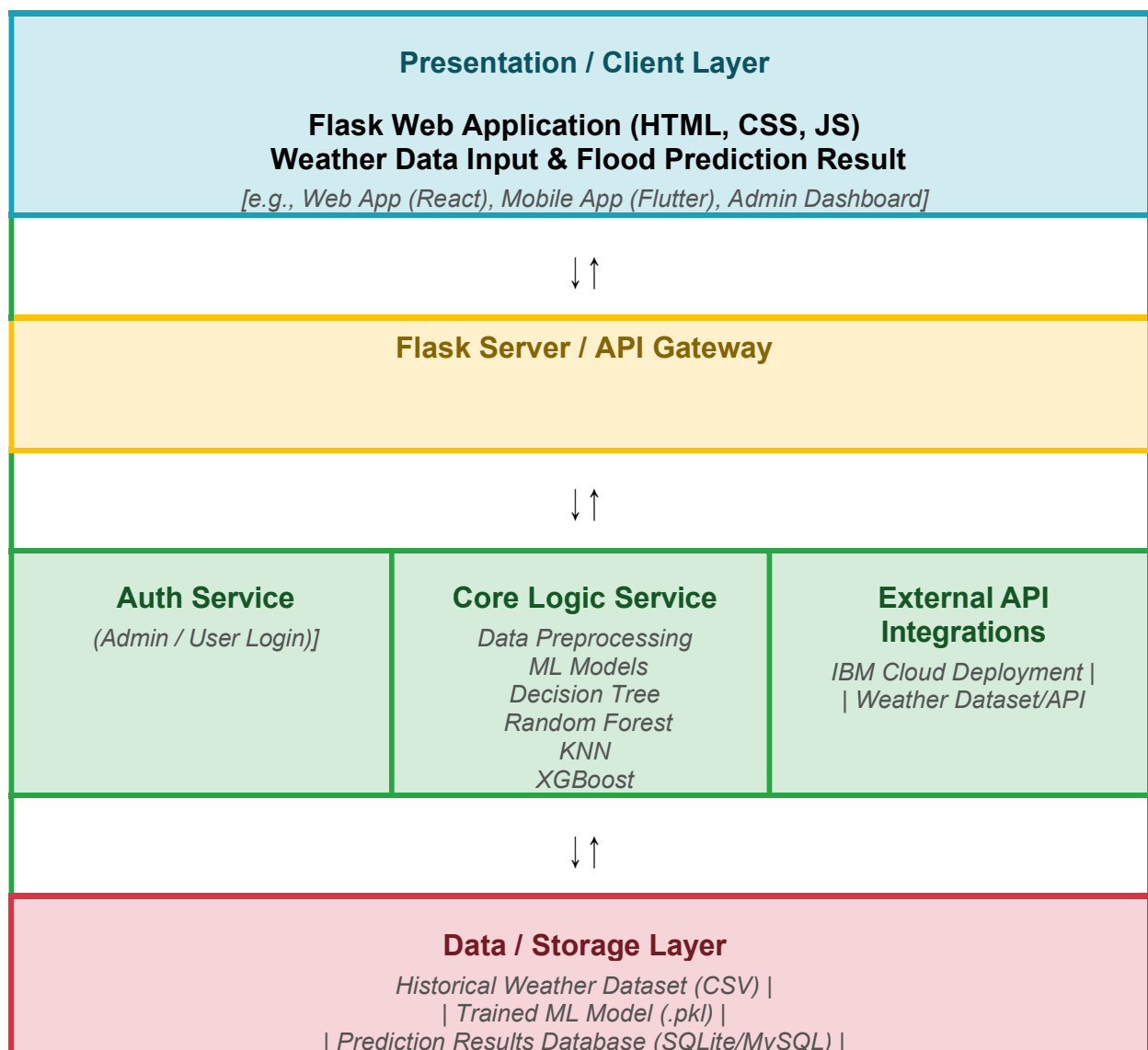


Solution Architecture

Date	2 JULY 2026
Team ID	XXXXXX
Project Name	Flood Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	<i>Provides a user-friendly interface for entering prediction results.</i>	<i>HTML, CSS, JavaScript, Flask</i>
<i>[API Gateway]</i>	<i>Receives user requests, validates inputs, and forwards data to the Machine Learning model.</i>	<i>Flask, Python</i>
<i>[Microservices]</i>	<i>Preprocesses weather data, executes prediction using trained ML models, and returns flood risk results.]</i>	<i>Python, Scikit-learn, XGBoost, Pandas, NumPy</i>
<i>[Database]</i>	<i>Stores historical weather datasets, trained machine learning models, and prediction records.</i>	<i>CSV Dataset, SQLite/MySQL, Pickle</i>